Name: _____

Class: _____Date:_____

# LESSON 16. MULTIPLICATION OF INTEGERS

3.26. Calculate the product 127 · 6. Then use it to find the following products:

a) (−127) · 6; b) 127 · (−6); c) (−127) · (−6).

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3.27. Without calculating, compare each product with 0:

a) 413 · 628; b) (−517) · 846; c) (−394) · (−725).

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3.28. Compare:

a) (+27) · (−15) and (−6) · (−9); b) (−36) · (−5) and (−12) · (−15);

c) (−18) · (+35) and (+28) · (−24).

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3.29. Let a be a negative integer. Is b a positive or a negative integer if:

a) the product a · b is a positive integer? b) the product a · b is a negative integer?

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3.30. Replace each "?" with a suitable number in the table below:

| x | −34 | 45 | −24 | −25 | 0 | −427 | −1 | −615 |
|---|---|---|---|---|---|---|---|---|
| y | 12 | −9 | −35 | −160 | −718 | 1 | 386 | −1 |
| x · y | ? | ? | ? | ? | ? | ? | ? | ? |

3.31. Find the integer x, given:

a) $7 \cdot (x + 35) = 0$; b) $(19 - x) \cdot (x + 14) = 0$; c) $(-x) \cdot (x - 52) = 0$.

3.32. Calculate in a smart way:

a) $(37 - 12) \cdot (-7) + (-16 - 9) \cdot 33$;

b) $(-148) \cdot (135 - 284) - 135 \cdot (284 - 148)$.

3.33. A clothing factory is changing to a new style of clothes. The fabric needed for each new outfit changes by x (dm) compared with the old style. In each case below, by how many decimetres does the fabric needed for 360 new outfits change?

a) $x = 15$; b) $x = -6$.

3.34. Seven integers have this property: the product of any three of them is always a negative integer. Is the product of all seven integers negative or positive? Explain why.